

An End-to-End Quality Assurance Approach for Web, Mobile, and API Platforms

Loyalty+ Store (Real Targets)

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01 Use case Overview

Problem Identification

- Ensuring correct system behavior.
- Identifying hidden functional and logical defects.
- Detecting performance limitations.

Project Goal

To utilize a multi-layered testing approach needed to strength software reliability.

Targeted Platforms

- SauceDemo Website
- DummyJSON API
- HyperMax Application

Project Objectives

- Design structured manual test cases and bug reports.
- Implement automated tests
- Validate API endpoints' reliability and performance

Testing Methods

Manual Testing, API Testing, Performance Testing, Web Automation Testing, and Mobile Automation Testing

2.1 Manual Testing

How?

Exploratory Test + Microsoft office Excel

Why?

30 Test Cases

9 Bug Reports (e.g., [Item14_BugReport.xlsx](#))

Verify Adding Items to Cart (2).	1- Navigate to (Products) page after logging in. 2- Click "Add to Cart" for specific items. 3- Observe the cart icon count after each addition.	Item Name: Sauce Labs Fleece Jacket.	The items should be added to the cart and the cart icon should display the updated number of items added.	The item were not added to the cart, and the cart icon count remained unchanged.	Fail	https://2cm.es/1hS9s	Log In Data: UserName: problem_user Password: secret_sauce Applicable Items (same behavior): 1. Test.allTheThings() T-Shirt (Red) 2. Sauce Labs Bolt T-Shirt
Verify Checkout – Filling in Required User Information.	1- Navigate to (Your Cart) page with items added. 2- Click the "Checkout" button. 3- On the (Checkout: Your Information) page, fill in the required fields. 4- Click the "Continue" button.	First Name: Hala, Last Name: Hanaqtah, Zip Code: 11196.	The system should accept the entered information and navigate to the (Checkout: Overview) page displaying selected items information and payment, shipping information as well as total price.	As Expected.	Pass	https://2cm.es/1hS85	Log In Data: UserName: standard_user Password: secret_sauce

2.2 Postman API Testing

How?

Postman was used on endpoints products, carts, authentication, and users.
Newman for reports ([report.html](#)).

Using

Environment and
Collection Variables

Data-driven with
CSV File

Positive and
Negative Scenarios

UserFirstName	UserLastName	UserAge	UserEmail	Search_Name	Invalid_Search	Valid_Id	Invalid_Id
Jhon	Doe	22	Jhon.Doe22@gmail.com	Emily	sleep	1	2000
Demo	User	25	Demo.User25@gmail.com	Emily	jump	2	500
Jane	Smith	20	Jane.Smith25@gmail.com	Emily	zzzz	3	300

Why?

Data Integrity,
Reliability, and
Error Handling

DummyJasonFinalProject + DummyJsonEnv

DummyJasonFinalProject Copy - Run results

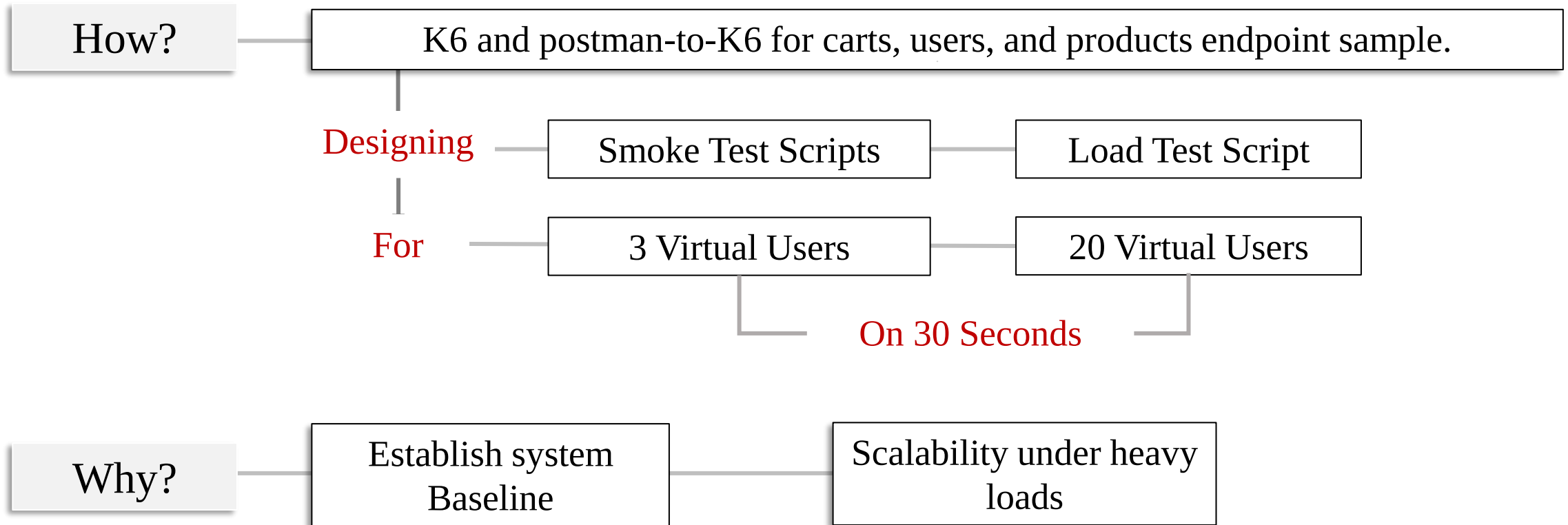
Debug with AI Run Again + New Run Automate Run Share

Ran today at 09:53:07 PM · View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	Dummy.JsonEnv	1	5s 637ms	118	0	85 ms

All Tests Passed (116) Failed (2) Skipped (0) Errors (0) Console Log View Summary

2.3 K6 Performance Testing



Smoke Test Results

```
THRESHOLDS
http_req_duration
✓ 'p(90)<250' p(90)=162.85ms
✓ 'p(95)<400' p(95)=301.5ms

http_req_failed
✓ 'rate<0.01' rate=0.00%

TOTAL RESULTS
checks_total.....: 204      5.316043/s
checks_succeeded...: 100.00% 204 out of 204
checks_failed.....: 0.00%  0 out of 204

✓ 200 Status Code
✓ content-type = application/json
✓ carts property exists
✓ carts array length > 0
✓ response is json obj
✓ All carts have required keys
✓ Content-Type = application/json

HTTP
http_req_duration.....: avg=168.42ms min=147.43ms med=156.22ms max=304.33ms p(90)=162.85ms p(95)=301.5ms
{ expected_response:true }...: avg=168.42ms min=147.43ms med=156.22ms max=304.33ms p(90)=162.85ms p(95)=301.5ms
http_req_failed.....: 0.00% 0 out of 36
http_reqs.....: 36      0.938125/s

EXECUTION
iteration_duration.....: avg=9.55s      min=9.47s      med=9.49s      max=9.74s      p(90)=9.73s      p(95)=9.74s
iterations.....: 12      0.312708/s
vus.....: 3      min=3      max=3
vus_max.....: 3      min=3      max=3

NETWORK
data_received.....: 469 kB 12 kB/s
data_sent.....: 9.6 kB 251 B/s
```

Load Test Results

```
THRESHOLDS
http_req_duration
✓ 'p(90)<550' p(90)=463.97ms
✓ 'p(95)<600' p(95)=523.86ms

http_req_failed
✓ 'rate<0.1' rate=0.00%

TOTAL RESULTS
checks_total.....: 1280      32.556703/s
checks_succeeded...: 100.00% 1280 out of 1280
checks_failed.....: 0.00%  0 out of 1280

✓ 200 Status Code
✓ Content-Type = application/json
✓ Response is an object
✓ carts property exists and is an array

✓ Valid key Types
✓ ReturnedID = RequestedID
✓ Returned Response is an Object
✓ Returned Response is Not Empty
✓ First Name has the Searched Word

HTTP
http_req_duration.....: avg=245.64ms min=151.14ms med=170.16ms max=665.45ms p(90)=463.97ms p(95)=523.86ms
{ expected_response:true }...: avg=245.64ms min=151.14ms med=170.16ms max=665.45ms p(90)=463.97ms p(95)=523.86ms
http_req_failed.....: 0.00% 0 out of 240
http_reqs.....: 240      6.104382/s

EXECUTION
iteration_duration.....: avg=9.77s      min=9.48s      med=9.65s      max=10.24s      p(90)=10.23s      p(95)=10.23s
iterations.....: 80      2.034794/s
vus.....: 18      min=18      max=20
vus_max.....: 20      min=20      max=20

NETWORK
data_received.....: 3.1 MB 80 kB/s
data_sent.....: 65 kB 1.6 kB/s

Running (0m39.3s), 00/20 VUs, 80 complete and 0 interrupted iterations
Default ✓ [=====] 20 VUs 30s
```

2.4 Web Automation Testing

How?

Selenium and TestNG on Standard User and Problem User

Including 11 Test Cases

Authentication

Browse Item

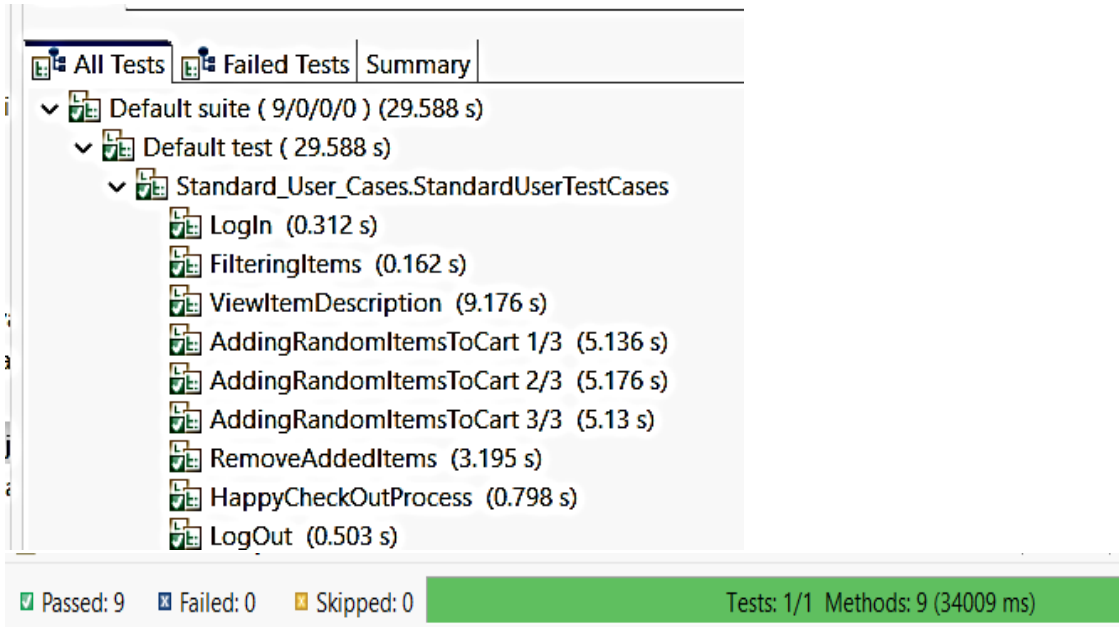
Item Sorting

Cart Management

Check out Process

Why?

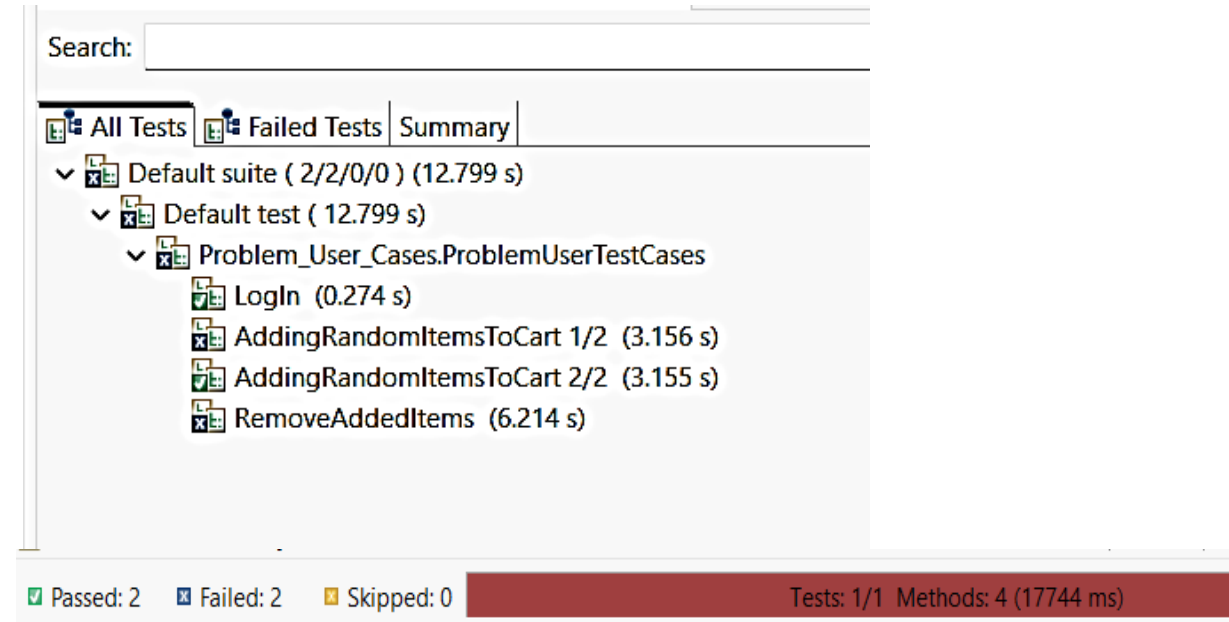
Execute and confirm the alignment between the expected and actual results



TestNG Results for Standard User Test Cases:

- All Tests | Failed Tests | Summary
- Default suite (9/0/0/0) (29.588 s)
 - Default test (29.588 s)
 - Standard_User_Cases.StandardUserTestCases
 - LogIn (0.312 s)
 - FilteringItems (0.162 s)
 - ViewItemDescription (9.176 s)
 - AddingRandomItemsToCart 1/3 (5.136 s)
 - AddingRandomItemsToCart 2/3 (5.176 s)
 - AddingRandomItemsToCart 3/3 (5.13 s)
 - RemoveAddedItems (3.195 s)
 - HappyCheckOutProcess (0.798 s)
 - LogOut (0.503 s)

Passed: 9 Failed: 0 Skipped: 0 Tests: 1/1 Methods: 9 (34009 ms)



TestNG Results for Problem User Test Cases:

- Search:
- All Tests | Failed Tests | Summary
- Default suite (2/2/0/0) (12.799 s)
 - Default test (12.799 s)
 - Problem_User_Cases.ProblemUserTestCases
 - LogIn (0.274 s)
 - AddingRandomItemsToCart 1/2 (3.156 s)
 - AddingRandomItemsToCart 2/2 (3.155 s)
 - RemoveAddedItems (6.214 s)

Passed: 2 Failed: 2 Skipped: 0 Tests: 1/1 Methods: 4 (17744 ms)

2.5 Mobile Automation Testing

How?

Appium, Appium Inspector, Android Emulator (PIXEL 5)

Asserting
through
Dedicated Test Script

Country
Selection

Category
Selection

Adding
Items

Item
Quantity

Error
Handling

Why?

Ensuring a seamless
native mobile
experience, from
initial onboarding to
cart management,
matching the
expectations.

```
2026-02-06T12:18:02.397Z INFO webdriver: COMMAND isElementDisplayed("00000000-0000-009e-ffff-ffff0000076b")
2026-02-06T12:18:02.398Z INFO webdriver: [GET] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element/00000000-0000-009e-ffff-ffff0000076b/displayed
2026-02-06T12:18:02.424Z INFO webdriver: RESULT true
2026-02-06T12:18:02.424Z INFO webdriver: COMMAND getElementText("00000000-0000-009e-ffff-ffff0000076b")
2026-02-06T12:18:02.425Z INFO webdriver: [GET] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element/00000000-0000-009e-ffff-ffff0000076b/text
2026-02-06T12:18:02.442Z INFO webdriver: RESULT Alwadi Tender Chicken Breast 2 Kg
2026-02-06T12:18:03.444Z INFO webdriver: COMMAND findElement("accessibility id", "AddToCartButton")
2026-02-06T12:18:03.445Z INFO webdriver: [POST] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element
2026-02-06T12:18:03.445Z INFO webdriver: DATA {"using":"accessibility id","value":"AddToCartButton"}
2026-02-06T12:18:03.575Z INFO webdriver: RESULT {
  'element-6066-11e4-a52e-4f735466cecf': '00000000-0000-009e-ffff-ffff00000874',
  ELEMENT: '00000000-0000-009e-ffff-ffff00000874'
}

2026-02-06T12:18:14.500Z INFO webdriver: COMMAND findElement("accessibility id", "SelectQuantity")
2026-02-06T12:18:14.500Z INFO webdriver: [POST] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element
2026-02-06T12:18:14.501Z INFO webdriver: DATA {"using":"accessibility id","value":"SelectQuantity"}
2026-02-06T12:18:14.876Z INFO webdriver: RESULT {
  'element-6066-11e4-a52e-4f735466cecf': '00000000-0000-009f-ffff-ffff00000b46',
  ELEMENT: '00000000-0000-009f-ffff-ffff00000b46'
}

2026-02-06T12:18:16.890Z INFO webdriver: COMMAND getElementText("00000000-0000-009f-ffff-ffff00000b46")
2026-02-06T12:18:16.891Z INFO webdriver: [GET] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element/00000000-0000-009f-ffff-ffff00000b46/text
2026-02-06T12:18:16.914Z INFO webdriver: RESULT 2 Pieces
2026-02-06T12:18:16.915Z INFO webdriver: COMMAND elementClick("00000000-0000-009f-ffff-ffff00000b46")
2026-02-06T12:18:16.915Z INFO webdriver: [POST] http://127.0.0.1:4723/session/43bb9583-5bdf-4b80-a744-45e4e61244f8/element/00000000-0000-009f-ffff-ffff00000b46/click
```

03 CI/CD Integration

CI/CD pipeline was implemented to automate test execution and ensure continuous validation

← My project deploy

✓ Final Project Documentation #74

Summary

All jobs

✓ deploy

Run details

Usage

Workflow file

Triggered via push 2 hours ago

Hanaqtah pushed

2a42b01

main

Status

Success

Total duration

2m 7s

Artifacts

–

deploy.yml

on: push

✓ deploy

1m 46s

deploy

succeeded 2 hours ago in 1m 46s

> ✓ Set up job

> ✓ Update repository

> ✓ Newman Test

> ✓ K6 Smoke Test

> ✓ CPU status

> ✓ Ram status

> ✓ Complete job

04 Conclusion

- Integrating these robust testing strategies reinforces the value of automated quality gates in modern software delivery.
- It demonstrates how quality assurance can evolve from a final checkpoint into a continuous, scalable safety net. Since functional checks alone does not ensure system robustness, uncovering role-specific defects and performance limitations.

Treat quality as a system-wide responsibility—embed multi-layered testing, automation, performance validation, and CI/CD enforcement from day one to deliver resilient, scalable, and production-ready software.

**Thank you for your
attention.**